

IIAI30013

Computer Vision

HW4: Segmentation

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HW4: Segmentation

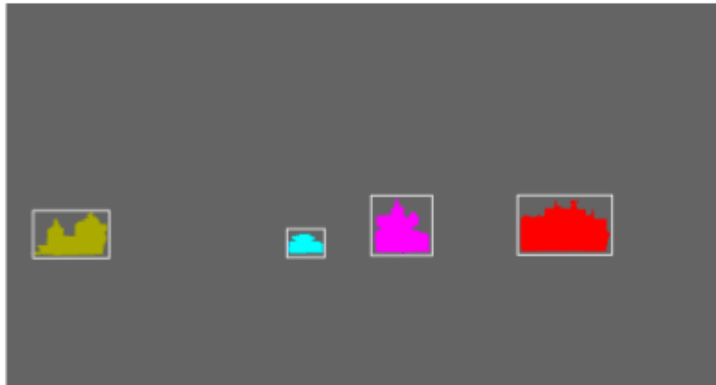
Image



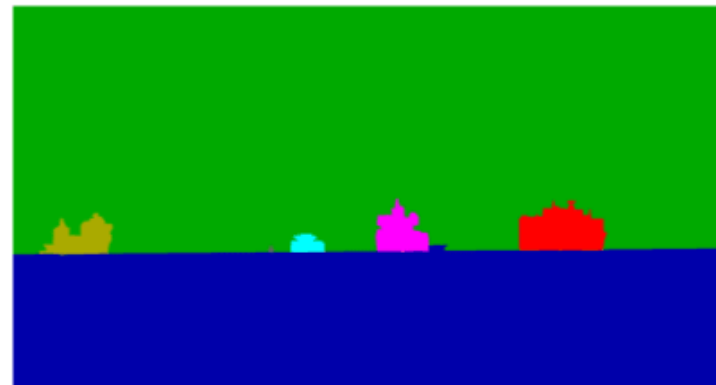
Semantic Segmentation



Instance Segmentation



Panoptic Segmentation



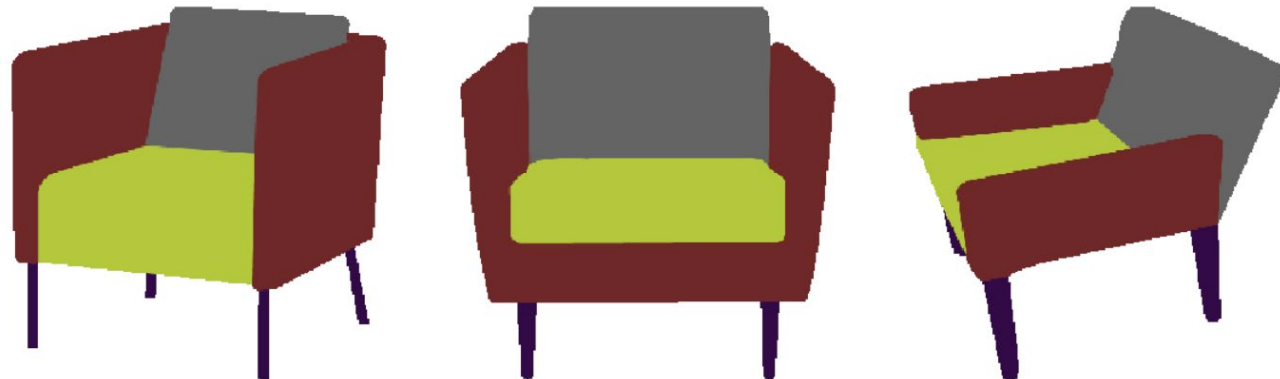
HW4: Segmentation

- Our homework this time is **Semantic Segmentation**:
 - Component Segmentation of Sofa

Input



Output



HW4: Segmentation

- Training Set:
 - 400 images with GT.

test

train

input

GT



HW4: Segmentation

- Test Set:
 - 10 images with GT.

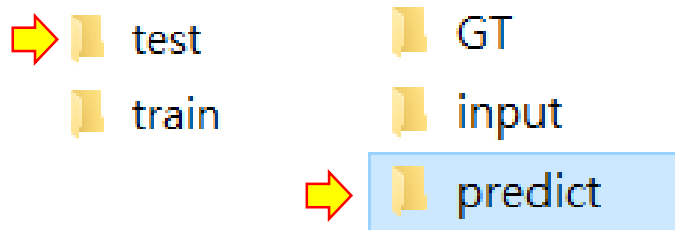


Please refrain from using ground truth data for training as it constitutes **cheating**.

HW4: Segmentation

- Submission:

- Submit 10 segmentation predictions generated by your trained model and save them to the "predict" folder.



HW4: Segmentation

- Homework due: 5/20 23:59
- Late submissions will incur a penalty of one point for each day overdue.
- The assignment allows a maximum extension of 3 days (it will not be accepted if submitted later than 3 days).
- Submit files:
 - 1) Code file (**.py** or **.ipynb**) containing your implementation.
 - 2) **3 images** of Validation results at epoch 5, 10, and 20.
 - 3) Test results consisting of **10 segmentation images**.
 - 4) Report (**pdf format**): include both quantitative and qualitative analysis of the validation and test results. Use PSNR and IoU as metrics for the quantitative analysis.
- This assignment can be carried out using [Colab](#) or completed on your PC.

HW4: Segmentation

- Example of Validation Result



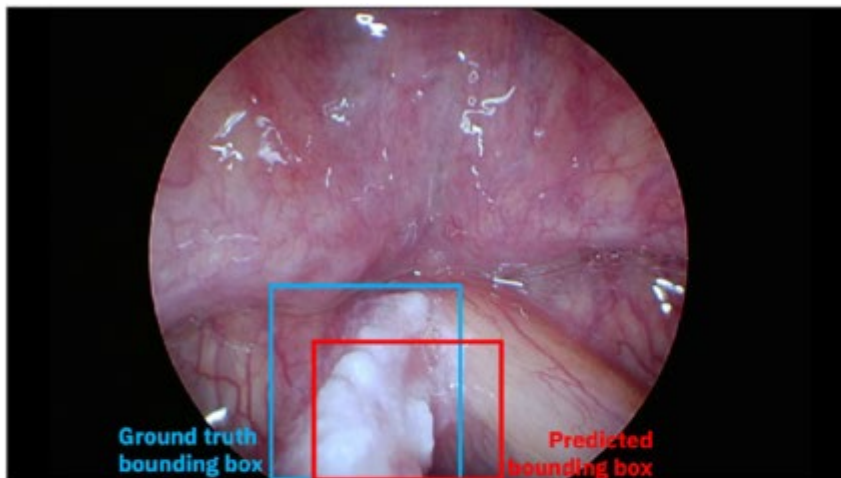
HW4: Segmentation

- Rules:

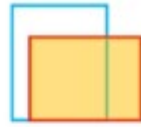
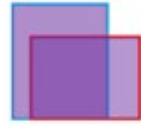
- (1) Participants are allowed to download pre-trained models for fine-tuning.
- (2) Participants may also train a new model from scratch for prediction.
- (3) Higher scores will be awarded to participants who build their models from scratch.
- (4) Implement an **IoU** function to calculate the IoU value for each segmentation results and the overall **mIoU**.
- (5) Implement an **PSNR** function to calculate the IoU value for each segmentation results and the overall **mean of PSNR**.

HW4: Segmentation

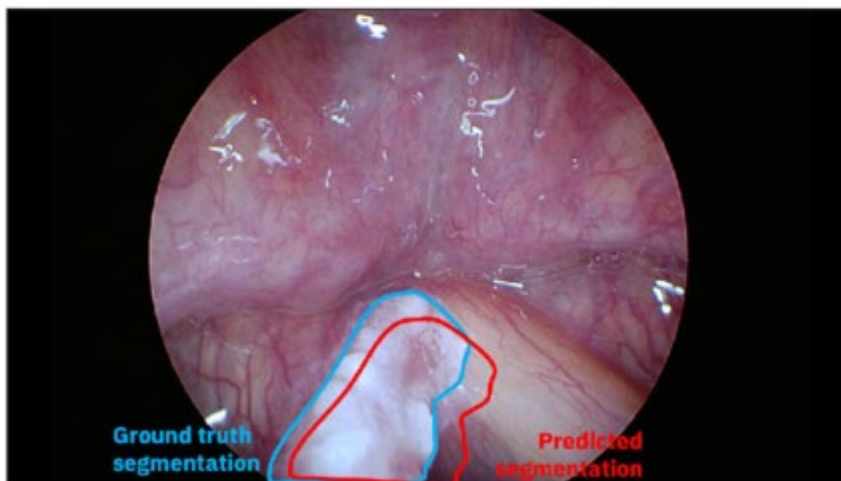
- Evaluation Metrics – IoU



Detection


$$\text{IoU} = \frac{\text{area of overlap}}{\text{total area of union}}$$



Bounding Box



Segmentation


$$\text{IoU} = \frac{\text{area of overlap}}{\text{total area of union}}$$



Mask

HW4: Segmentation

• Example

- PSNR: 17.71
- IoU: Armrest (brown): 97.13%, Leg (black): 95.99%,
Seat Base (yellow): 85.40%, Backrest (gray): 97.28%

- Cushion: (60, 180, 90) – green
- Armrest: (110, 40, 40) – brown
- Leg: (50, 10, 70) – black
- Seat Base: (180, 200, 60) – yellow
- Backrest: (100, 100, 100) – gray

```
# select random example
ix = randint(0, len(X1), 1)
src_image, tar_image = X1[ix], X2[ix]
gen_image = model.predict(src_image)
print(ix, PSNR(gen_image, tar_image)[0])
plot_images(src_image, gen_image, tar_image)
```

[30] 17.713072



```
r=60;g=180;b=90
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅墊:",result[1])
```

```
r=110;g=40;b=40
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("扶手:",result[1])
```

```
r=50;g=10;b=70
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅腳:",result[1])
```

```
r=180;g=200;b=60
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅子底:",result[1])
```

```
r=100;g=100;b=100
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅墊:",result[1])
```

```
椅墊: 0
扶手: 0.9713475751746482
椅腳: 0.9599682392642096
椅子底: 0.8546839299314547
椅墊: 0.9728260869565217
```

HW4: Segmentation

• Example

- PSNR: 17.21
- IoU: Leg (black): 77.18%

Seat Base (yellow): 98.49%

- Cushion: (60, 180, 90) – green
- Armrest: (110, 40, 40) – brown
- Leg: (50, 10, 70) – black
- Seat base: (180, 200, 60) – yellow
- Backrest: (100, 100, 100) – gray

```
# select random example
ix = randint(0, len(X1), 1)
src_image, tar_image = X1[ix], X2[ix]
gen_image = model.predict(src_image)
print(ix, PSNR(gen_image, tar_image)[0])
plot_images(src_image, gen_image, tar_image)
```

[1002] 17.214153



```
r=60;g=180;b=90
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅墊:",result[1])
```

```
r=110;g=40;b=40
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("扶手:",result[1])
```

```
r=50;g=10;b=70
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅腳:",result[1])
```

```
r=180;g=200;b=60
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅子底:",result[1])
```

```
r=100;g=100;b=100
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅墊:",result[1])
```

```
椅墊: 0
扶手: 0
椅腳: 0.771875
椅子底: 0.9849449088145896
椅墊: 0
```

HW4: Segmentation

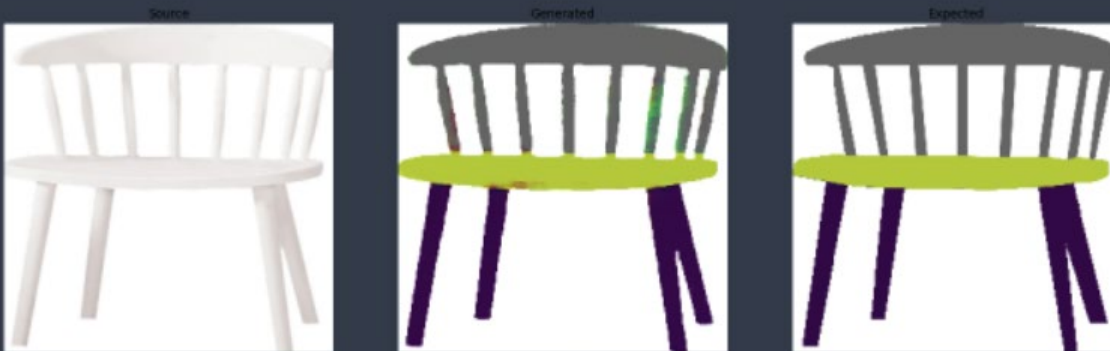
• Example

- PSNR: 11.40
- IoU: Leg (black): 95.80%, Seat Base (yellow): 90.30%,
Backrest (green): 94.92%

- Cushion: (60, 180, 90) – green
- Armrest: (110, 40, 40) – brown
- Leg: (50, 10, 70) – black
- Seat base: (180, 200, 60) – yellow
- Backrest: (100, 100, 100) – gray

```
# select random example
ix = randint(0, len(X1), 1)
src_image, tar_image = X1[ix], X2[ix]
gen_image = model.predict(src_image)
print(ix, PSNR(gen_image, tar_image)[0])
plot_images(src_image, gen_image, tar_image)
```

[624] 11.400866



```
r=60;g=180;b=90
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅墊:",result[1])
```

```
r=110;g=40;b=40
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("扶手:",result[1])
```

```
r=50;g=10;b=70
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅腳:",result[1])
```

```
r=180;g=200;b=60
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅子底:",result[1])
```

```
r=100;g=100;b=100
cnt=0;s_iou=0
result=iou(r,g,b,gen_image,tar_image)
print("椅墊:",result[1])
```

```
椅墊: 0
扶手: 0
椅腳: 0.9580444273240883
椅子底: 0.9030664147652957
椅墊: 0.94921875
```

HW4: Segmentation

- Result

- Cushion (green): 76.76%

```

椅墊: (60,180,90) (綠色)
In [31]: r=60;g=180;b=90
         cnt=0;s_iou=0
         for k in test:
             k=np.array([k])
             src_image, tar_image = X1[k], X2[k]
             gen_image = model.predict(src_image)
             result=iou(r,g,b,gen_image,tar_image)
             cnt+=result[0]
             s_iou+=result[1]

         s_iou/cnt

0.7676898467979737

```

- Armrest (brown): 77.64%

```

扶手: (110,40,40) (咖啡色)
In [32]: r=110;g=40;b=40
         cnt=0;s_iou=0
         for k in test:
             k=np.array([k])
             src_image, tar_image = X1[k], X2[k]
             gen_image = model.predict(src_image)
             result=iou(r,g,b,gen_image,tar_image)
             cnt+=result[0]
             s_iou+=result[1]

         s_iou/cnt

0.776427139455019

```

HW4: Segmentation

- Result

- Leg (black): 67.86%

```

椅腳: (50,10,70) (黑色)

In [33]: r=50;g=10;b=70
          cnt=0;s_iou=0
          for k in test:
              k=np.array([k])
              src_image, tar_image = X1[k], X2[k]
              gen_image = model.predict(src_image)
              result=iou(r,g,b,gen_image,tar_image)
              cnt+=result[0]
              s_iou+=result[1]

          s_iou/cnt

0.6786286786968682

```

- Seat Base (yellow): 75.83%

```

椅子底: (180,200,60) (黃色)

In [34]: r=180;g=200;b=60
          cnt=0;s_iou=0
          for k in test:
              k=np.array([k])
              src_image, tar_image = X1[k], X2[k]
              gen_image = model.predict(src_image)
              result=iou(r,g,b,gen_image,tar_image)
              cnt+=result[0]
              s_iou+=result[1]

          s_iou/cnt

0.7583794512032871

```

HW4: Segmentation

- Result

- Backrest (gray): 60.20%

```

椅背: (100,100,100) (灰色)

In [35]: r=100;g=100;b=100
          cnt=0;s_iou=0
          for k in test:
              k=np.array([k])
              src_image, tar_image = X1[k], X2[k]
              gen_image = model.predict(src_image)
              result=iou(r,g,b,gen_image,tar_image)
              cnt+=result[0]
              s_iou+=result[1]

          s_iou/cnt

0.6020309908219154

```

- Overall IoU: 71.66%

```

(0.7676898467979737+0.776427139455019+0.6786286786968682+0.7583794512032871+0.6020309908219154)/5

0.7166312213950127

```

IIAI30013

Computer Vision

HW4: Segmentation

Q & A

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